

Limited-Memory Renewal Contracts: Exact Quadratic Design and Prefix Decomposition

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Abstract. A renewal provider must allocate accepted obligations through a limited execution alphabet. We study joint alphabet and promise design with local participation limits, bounded realization exposure, heterogeneous shortfall costs, and charges for enabled levels. An eligibility-prefix reduction gives the exact response to any branch promise. Our main algorithmic result converts the price relaxation into an ordered path problem, solved in polynomial time with variable branch count and alphabet budget. The same decomposition supplies completion bounds for finite global search and independently checkable intervals when search stops early. For rational quadratic primitives, we retain an exact unrestricted continuous reference algorithm. Feasibility-preserving rounding connects catalog bounds to continuous values, and a fixed-input lower bound proves the uniform mesh rate sharp. The sharper saturated theory retains cap anchors, an off-cap highest level, and linear-time dynamic-programming layers after contractual tail minimization. A monotone-charge theorem separates a genuine institutional boundary from a specially shaped randomization example. Synthetic computations compare exact rational methods, report enumeration and arithmetic growth, and cross-check a separate original-policy nonlinear formulation. The decomposition extends to an abstract ordered-interface allocation class without requiring the renewal interpretation.

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1 Introduction

An upstream renewal desk can know an accepted obligation that a downstream execution gateway does not retain. When only a small service symbol crosses that boundary, the provider must choose both its terminal instruction alphabet and the branch-dependent encoder. This is a quantizer-design problem, but its objective and feasible randomization are generated by a contract: an aggregate continuation promise, local participation limits, and the cost of intermediate rather than terminal service. We ask how to choose the alphabet and allocate the promise jointly when individual lottery realizations have an overrun limit and enabling an additional execution level has a cost.

Our principal algorithmic result uses this contractual structure rather than enumerating all its possible local decisions. After assigning a price to the aggregate promise, an optimal ordered alphabet can be represented by a path whose edges account for the branch caps they cross. A dynamic program solves this price problem in polynomial time, with both the number of branches and the alphabet budget allowed to vary. The same decomposition gives exact bounds for partially chosen alphabets. A finite global search then returns either an exact catalog optimum or an explicit lower and upper bound at interruption. Its witness records the complete search partition and can be checked independently of the search code. No equality between the price relaxation and the original problem is presumed.

The method is part of a unified theory covering every feasible aggregate promise, continuously chosen execution levels, bounded realization exposure, and charges on the actual selected levels. A branch-specific eligibility prefix and a canonical adjacent lottery reduce each fixed-book problem to concave resource allocation. The general concave model admits this exact fixed-book structure, the priced decomposition, and a feasible discretization bound; exact unrestricted continuous optimization is proved for the rational quadratic subclass. For that subclass, eliminating lottery probabilities gives closed quadratic cells, and exhaustive stationary-face search supplies a finite exact reference algorithm, including collisions and nonsaturated pathwise participation. We retain that result and state its full exponential cost.

A second connection turns a numerical catalog into a controlled approximation of continuous design. Downward rounding and reconstructed lottery probabilities preserve every branch promise and realization ceiling exactly, even at binding constraints. A uniform error bound converts the search interval into a continuous-value interval, separating incomplete optimization from discretization. We prove that its linear mesh rate is order-sharp on a single fixed, uncharged input. The result does not require catalog optimization to finish, and it does not claim an efficient approximation scheme for unrestricted alphabet budgets.

The saturated contract remains especially tractable. Deterministic execution gives an ordered one-sided quantizer. Under expected participation, a quadratic optimum has cap-anchored nonhighest levels and a continuously optimized highest level that can lie between caps. A crossing-difference identity preserves Monge order after continuous tail minimization, allowing classical linear work per dynamic-programming layer. We retain those results and proofs, the all-promise exact-alphabet phase diagram, stability bounds, and the faster saturated charged-catalog recurrence. The new decomposition broadens the algorithmic treatment without replacing these sharper endpoint results.

Ordered quantizer dynamic programming, optimized reproduction points, unbiased splitting, Lagrangian relaxation, and matrix search are established ideas (Wu, 1991; Pagès and Wilbertz, 2012; Croci et al., 2022;

Jourdain and Pagès, 2021; Handler and Zang, 1980). The distinct result is how exact promises, heterogeneous shortfall costs, support eligibility, and a shared selected alphabet produce a polynomial priced path problem and reusable completion bounds. We formulate this result for an ordered-interface allocation class that does not require the renewal interpretation. The operational model remains an explicitly stipulated risk-neutral or actuarial acceptance rule followed by restricted execution; it is not presented as customer data or an implemented field deployment. Synthetic studies test exact bounds, expose the continuous reference solver’s enumeration frontier, and compare a separate original-policy nonlinear formulation. A monotone-charge theorem also identifies precisely why the retained one-branch randomization example depends on its specially shaped nonmonotone charge.

2 Quantization, Contracting, and the Contribution Boundary

Ordered scalar quantization. Wu (1991) accelerates the discrete optimal mean-square K -level quantization dynamic program from $O(KN^2)$ to $O(KN)$ by matrix searching. Later one-dimensional clustering treatments extend the cost structures and discuss reconstruction and space reductions (Grønlund et al., 2017). Thus contiguous-cell dynamic programming, continuous reproduction-point optimization, and an $O(mk)$ matrix-search consequence are not new as generic claims. In the present deterministic problem, participation forces a cell’s reconstruction to its smallest cap, not its least-squares centroid. Its loss includes branch-specific intermediate-service costs. The difference is a particular feasible distortion, not a new ordered-partition paradigm.

Unbiased randomization and dual quantization. The adjacent lottery is exactly the classical unbiased two-point quantizer. For distinct adjacent levels $u < v$ and $u \leq b \leq v$,

$$\Pr(C = v) = \frac{b - u}{v - u}, \quad \Pr(C = u) = \frac{v - b}{v - u}, \quad \mathbb{E}C = b. \quad (1)$$

Stochastic rounding uses this same rule (Crocì et al., 2022). Dual quantization is an even closer optimization predecessor: Pagès and Wilbertz (2012) optimize mean-preserving random splitting and relate it to piecewise-affine interpolation. Their extended formulation also uses nearest-neighbor projection outside the grid’s convex hull. We therefore do not claim that random splitting, optimized grids, interpolation, or treatment of out-of-hull points first arise here.

For the quadratic terminal reward, our in-hull loss is $q(b - u)(v - b)/2$, the corresponding quadratic dual-quantization error multiplied by $q/2$. Above the highest level z , however, the contractual loss is

$$f(b) - f(z) + h_j(b - z) = (r - qz)(b - z) + \frac{\gamma_j - q}{2}(b - z)^2.$$

It is generally not a common squared-distance projection penalty: it carries a level-dependent linear term and a heterogeneous intermediate-service curvature. The paper’s combined saturated structural result concerns this one-sided contractual tail, its continuously chosen endpoint, and the cancellation that preserves the terminal Monge order. The affine movement of an interior codeword follows the quadratic interpolation

structure and is not advertised as an independent general quantization principle. Its sufficient conditions are explicit: saturated branch targets, quadratic terminal reward, uncharged codewords, and intermediate costs arising only above the highest level. Section 8 treats endogenous targets, and Section 12 treats actual level charges without assuming the same anchoring lemma.

Optimized dual grids and enabled-level penalties. Jourdain and Pagès (2021) prove uniqueness of optimal one-dimensional dual grids for compactly supported log-concave densities and develop a Lloyd-like algorithm in the quadratic case. That result directly concerns optimized grids, not merely evaluation of a prescribed rounding rule. Here branch targets are themselves constrained decisions, intermediate shortfall costs can differ by branch, and each branch admits a different eligible prefix. Our exact continuous reduction does not assume a log-concave source density or uniqueness of a stationary grid; the grid approximation also preserves the contractual constraints at their boundary.

A charge for enabling a representative is a fixed-opening-cost idea, not a new type of resource pricing. For example, Aardal et al. (2014) distinguish facility-opening costs from demand-dependent service costs under a cardinality restriction. In our model, a selected level can implement adjacent random splitting and branch-specific intermediate shortfalls, rather than serving a client by ordinary distance to one center. The opening charge is not entropy, expected message length, or a cost per token use. The new issue is its joint optimization with exact promises and realization eligibility. The finite quadratic search in Theorem 9.3 invokes a classical optimization principle; its contribution is the contractual reduction to those closed cells. Theorem 11.1 gives the separate, constraint-preserving approximation guarantee.

Finite-rate control and contracts. An informed encoder and a downstream finite-alphabet decoder are standard in communication-constrained control (Fu, 2024). Our result is not a stabilization or channel-capacity theorem. There is no dynamic plant whose state is estimated through noisy feedback, no hidden type report, and no incentive-compatibility claim. What the contract adds is the allocation of accepted continuation payments and the distinction between conditional expected and realization-wise participation. Continuation promises themselves have a long contracting history (Spear and Srivastava, 1987; Thomas and Worrall, 1988), including finite policy-graph implementations (Zhang, 2012). Below the exact alphabet threshold, participation determines the allowable quantizer rather than communication alone determining the objective.

Classical matrix search, model-specific cost structure. Quadrangle-inequality dynamic programming (Yao, 1980) and full totally monotone matrix search (Aggarwal et al., 1987) supply the algorithmic framework. Generalized and partial Monge matrix searching require their own domain analysis (Klawe and Kleitman, 1990; Chan, 2021; Kaplan et al., 2017). Our suffix and prefix domains have the favorable completion orientation; this is not an improvement on the general partial-matrix bounds. We retain complete proofs of the finite tie rule, wholly padded selected rows, column offsets, and reachable predecessor intervals so that the implementation’s use of the classical result is checkable. The essential contractual input is the inequality after continuous tail minimization, not merely convexity of a scalar reproduction-point objective. We do not claim that minimizing a reproduction point preserves Monge structure for arbitrary quantizer losses.

Table 1 states the predecessor principle and contractual increment for each result.

3 The Contract and Its Execution Boundary

3.1 A saturated renewal agreement

At the renewal date, a public branch $j \in \{1, \dots, k\}$ is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. Its continuation cap is $0 < b_1 < \dots < b_k < 1$. The provider chooses an intermediate tier $Y_j \in [0, 1]$ and a terminal tier $C_j \in [0, 1]$. All branches subsequently reach the same terminal public vertex. Unit payment coefficients and unit discounting make total remaining payment $Y_j + C_j$. A fixed service-benefit continuation minus the customer's outside continuation gives the cap. At the root the provider has committed to

$$\sum_j \pi_j \mathbb{E}[Y_j + C_j] = \bar{b}, \quad \bar{b} := \sum_j \pi_j b_j. \quad (2)$$

Under expected participation, $\mathbb{E}[Y_j + C_j] \leq b_j$ for every j , with expectation conditional on that observed branch and taken before the internal draw. Positivity of the probabilities and (2) imply equality separately on every branch. Under pathwise participation, the stronger constraint is $Y_j + C_j \leq b_j$ almost surely. The same argument then implies equality almost surely, not merely in expectation.

The operating payoff is $f(C_j) - h_j(Y_j)$, with increasing concave f and convex nondecreasing h_j satisfying $h_j(0) = 0$. General shape assumptions suffice for the deterministic and pathwise results. The global continuous algorithm specializes to the quadratic primitives in (8). The negative intermediate contribution represents the net cost of an advance or expedited tier, while the terminal tier contributes net operating return. It does not mean that the customer has negative service benefit: that benefit has already entered the continuation cap. There are no endogenous public transitions, coupled commitments, or switching costs in the renewal theorem.

3.2 What a writable code means

At the observed branch, an encoder selects a symbol $Z \in \{1, \dots, m\}$. At the terminal vertex the decoder observes only the current public vertex and Z . Its stored program can contain arbitrary read-only model constants but cannot recover the realized branch from those constants. An inherited intermediate tier, a customer identifier with a branch lookup, a retained shared random seed, and a transmitted numerical payment are all additional history channels; none is free in this interface. An m -symbol alphabet requires $\lceil \log_2 m \rceil$ fixed-width writable bits, not m bits. Public-state storage and read-only program storage are counted separately.

Private encoder randomization is allowed under expected participation. Fresh decoder randomness is independent of earlier private choices conditional on the symbol. Any seed that correlates those choices across dates must itself be encoded in the charged symbol. Concavity permits a deterministic terminal decoder at the expected-participation optimum, so a codebook is a collection of terminal levels. The lower bound for exact implementation continues to allow randomized encoders and decoders under this accounting.

Distinct full-information terminal actions and strict concavity remain visible assumptions; Proposition 8.2 handles repeated caps; uniqueness is required for the exact-action lower bound.

3.3 Acceptance timing

The renewal desk observes the branch and offers the intermediate tier and terminal lottery. Under expected participation, the customer accepts before the ticket draw; after acceptance only the ticket reaches the common gateway. The realized ticket may be disclosed, but no new costless exit option is supplied after the draw within this episode. Under pathwise participation, each possible realized obligation must instead satisfy the branch cap, and a draw-specific intermediate tier is allowed. A seed known before acceptance changes that conditioning information. Repeated-customer effects require an enlarged state and are not part of the three-date theorem. The full stipulated mechanism and its qualifications are given in the electronic companion; no empirical or legal claim about an actual provider is made. Section 8 allows a nonsaturated root promise, and Section 12 specifies the intermediate risk protocol.

3.4 Preferences and a restricted execution interface

A concrete preference interpretation of expected participation is quasilinear, risk-neutral continuation utility $A_j - \mathbb{E}(Y + C)$ with outside value O_j , so that $b_j = A_j - O_j$ after normalization. Alternatively the expectation bound can be an explicitly agreed actuarial budget. Neither interpretation follows from a finite communication alphabet. A customer with a hard realization-level liquidity limit instead requires $Y + C \leq b_j$. The intermediate institution retains the actuarial cap and separately authorizes at most δ realization overrun. Its intermediate tier is committed before the draw so that the disclosed lottery and its ceiling can be checked together. Allowing a fresh, costless rejection after a high draw changes the feasible problem; the expected-participation comparison assumes that option is absent, not that every real renewal contract can eliminate it.

The downstream restriction is an interface restriction, not an inability to store a model. An upstream desk keeps the branch record; a downstream gateway has a read-only map from a small session token to tested execution levels and no access to the upstream record or a branch-indexed customer identifier. A relevant documented component analogy is operational-technology allowlisting and least functionality: NIST SP 800-82 Rev. 3 discusses restricting executable applications, testing policy configurations, disabling nonessential interfaces, and restricting writable set points (Stouffer et al., 2023, Sections 5.2.5.1, 6.2.4.1 and control CM-7). These are precedents for a constrained execution boundary, not evidence that a particular provider uses our entire renewal or lottery protocol.

Under this design, opening an additional level can incur a one-time cost of engineering, test fixtures, authorization, and configuration assurance. This is the modeled source of $\rho(c)$: it is paid if the level is enabled, independently of the frequency with which its token is used. Per-use charges and entropy-based communication charges would be different objective terms. An exogenous catalog represents levels already eligible for certification; a numerical catalog chosen by the operator instead calls for the approximation theorem in Section 11. The model does not assert that every read-only contract must be inaccessible

downstream; it quantifies the consequence when that access is deliberately excluded. No empirical adoption, calibration, or enforceability conclusion is inferred from the analogy.

4 The Deterministic Service-Code Frontier

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch j is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in $[0, 1]$. Branch j has cap $0 < b_1 < \dots < b_k < 1$, and the root promise is $\bar{b} = \sum_j \pi_j b_j$. At the intermediate date the reward is $-h_j(y)$, where h_j is convex, nondecreasing, and $h_j(0) = 0$. The common terminal reward $f(c)$ is continuous, concave, and strictly increasing on $[0, 1]$. The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

Theorem 4.1 (Heterogeneous renewal frontier). *The unique full-information policy is $y_j = 0$, $c_j = b_j$, with value $J = \sum_j \pi_j f(b_j)$. Exact implementation requires k symbols under strict cap ordering; repeated caps are covered by Proposition 8.2. With at most m deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell $i, \dots, j$, its terminal tier is b_i and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (3)$$

The exact minimum loss is $D_m = D(\min(m, k), k)$, where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (4)$$

and all other infeasible entries equal $+\infty$. With a precomputed cell-cost table this requires $O(mk^2)$ arithmetic operations. For quadratic f and quadratic h_j , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

The proof is in Electronic Companion Section EC.1.1.

This strengthens and includes the quadratic equal-probability frontier. With $f(c) = 2c - c^2/2$, $h_j(y) = y^2/2$, and $\pi_j = 1/k$, (3) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (5)$$

For $k = 8$ and $b_j = j/9$, one, two, and three symbols lose $119/162$, $5/18$, and $49/324$, respectively. Two writable bits permit four symbols, not three. The segmentation recurrence is standard; the contract-specific cell loss, ordered-cell sufficiency, and exact memory interpretation are the substantive statements.

5 When Randomization Improves a Restricted Memory Budget

The exact-optimum lower bound in Theorem 8.1 does not imply that randomization is useless below that alphabet size. The renewal family permits an explicit answer under the declared conditional-expectation participation institution. At an intermediate public branch, the controller may randomize its terminal symbol; it carries no uncharged seed or branch record. Participation is evaluated before that private draw. Terminal actions may depend only on the terminal public vertex and the retained symbol.

For terminal levels $c_1 < \dots < c_s$, let F_c be the linear interpolant through $(c_i, f(c_i))$. Since f is concave and strictly increasing, this interpolant is concave and strictly increasing on $[c_1, c_s]$; for $s = 1$ it is defined at its sole point. Retain the caps, probabilities, and convex intermediate costs of Theorem 4.1.

Theorem 5.1 (Optimal lotteries for a fixed terminal alphabet). *For a fixed set of terminal levels, a feasible randomized controller exists if and only if $c_1 \leq b_1$. Its optimum at branch j has deterministic intermediate tier and terminal mean*

$$\mu_j = \min\{b_j, c_s\}, \quad y_j = b_j - \mu_j.$$

It randomizes over at most the two adjacent terminal levels bracketing μ_j . The optimal loss with at most m writable terminal symbols is consequently

$$D_m^{\text{ran}} = \min_{1 \leq s \leq m, \substack{0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq b_1}} \sum_j \pi_j \{f(b_j) - F_c(\min\{b_j, c_s\}) + h_j((b_j - c_s)_+)\}. \quad (6)$$

The minimum allows coincident levels by deleting unused duplicates. It satisfies $D_m^{\text{ran}} \leq D_m$. Equality with the full-information optimum still requires k symbols when the full-information actions are distinct and rewards are strictly concave. Formula (6) optimizes the encoding for each continuous codebook; it is not a claim that the joint codebook problem is solved by the deterministic segmentation recurrence.

The proof is in Electronic Companion Section EC.1.2.

Proposition 5.2 (A strict, exactly solved two-symbol separation). *In the quadratic renewal model, take three equiprobable branches with caps $1/4, 1/2, 3/4$, $f(c) = 2c - c^2/2$, and $h_j(y) = y^2/2$. Then*

$$D_2 = \frac{1}{8}, \quad D_2^{\text{ran}} = \frac{1}{96}, \quad D_3 = D_3^{\text{ran}} = 0. \quad (7)$$

Two symbols thus reduce the randomized loss to one twelfth of the best deterministic loss, without reducing the three-symbol requirement for exact implementation.

The proof is in Electronic Companion Section EC.1.3.

The institutional qualification is substantive. On the middle branch, the high private draw delivers terminal payment $3/4$ against an expected-continuation cap $1/2$. The lottery is feasible when participation is checked before the private draw, as stipulated here, but not when that branch cap must hold for every private draw. We do not transfer the randomized frontier to the pathwise institution. The deterministic frontier and the exact-memory lower bound retain their original domains. This separation explains why a statement

about the minimum alphabet at the exact optimum cannot settle the economic value of randomization under a smaller alphabet.

6 The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem 5.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem 8.1.

Retain the three-date renewal architecture of Theorem 4.1, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (8)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

6.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (9)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma 6.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (6) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.1.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition 6.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

6.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (10)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (11)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma 6.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (12)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (13)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (14)$$

Theorem 6.2 (Global frontier and exact complexity). *Under (8), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (15)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing

complexity. A conservative bound is $O(mk^2S^3)$ bit operations with $S = O(k[L + \log(k+2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (6).

The proof is in Electronic Companion Section EC.1.5.

6.3 A continuous optimal highest level

Proposition 6.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation codebook is $(1/4, 5/8)$, and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (16)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.1.6.

6.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem 6.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem 4.1, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (17)$$

Consequently (15) and the deterministic frontier give an exact comparison of the two participation institutions under (8). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

The proof is in Electronic Companion Section EC.1.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem 6.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

7 Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

7.1 The inequality survives continuous terminal minimization

Write $I_\ell = [b_\ell, b_{\ell+1}]$, including $I_k = \{b_k\}$, and $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$ for $i \leq \ell$. A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, c < d, \quad (18)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

Lemma 7.1 (Contractual finite-cost inequalities). *The anchored edge costs E and deterministic cell costs C satisfy $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$ for $a \leq b \leq c \leq d$. The continuously minimized terminal array Q is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.1.8.

Lemma 7.2 (Previous-layer values are column offsets). *Adding any finite column term v_c preserves the Monge inequality. For fixed finite predecessor values, the arrays $A(j, i) = P(s, i) + E(i, j)$ and $B(j, i) = D(s - 1, i) + C(i + 1, j)$ are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.1.9.

7.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (19)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (20)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (20).

Lemma 7.3 (Suffix-feasible completion). *Let row r have feasible columns $c \geq \ell_r$, with nondecreasing cutoffs ℓ_r . Suppose the finite entries are Monge. Complete every $c < \ell_r$ with $K(r, c) = (1, 0, -c)$. Then (20) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.1.10.

Lemma 7.4 (Prefix-feasible completion). *Let row r have feasible columns $c \leq u_r$, with nondecreasing cutoffs u_r . Suppose the finite entries are Monge. Complete every $c > u_r$ with $K(r, c) = (1, 0, c)$. Then (20) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.1.11.

These lemmas specialize the known favorable completion setting in Section 2; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

7.3 Reachability and three search interfaces

Lemma 7.5 (Contiguous reachable predecessor sets). *For the anchored program, $P(1, j)$ is finite exactly at $j = 1$, and for $s \geq 2$, $P(s, j)$ is finite exactly when $s \leq j \leq k$. For the deterministic program, $D(0, j)$ is finite only at $j = 0$, and $D(s, j)$ is finite exactly when $s \leq j \leq k$ for $s \geq 1$. Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.1.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

Terminal interface. Rows and columns are $1, \dots, k$. At (i, ℓ) return $(1, 0, -\ell)$ if $\ell < i$; otherwise minimize (12) by (13) and return $(0, Q(i, \ell), \ell)$. Search all rows. Reevaluate each winning cell once to store T_i and its continuous optimizer.

Anchored-prefix interface. To form layer $s+1$, use rows $j = s+1, \dots, k$. Columns are $\{1\}$ for $s = 1$, and $s, \dots, k-1$ otherwise. At (j, i) return $(1, 0, i)$ if $i \geq j$, and $(0, P(s, i) + E(i, j), i)$ otherwise. Search and store values and predecessors. Scan the current reachable endpoints with T_i to update the at-most-budget incumbent; backtrack only its winning codebook.

Deterministic-prefix interface. To form layer s , use rows $j = s, \dots, k$. Columns are $\{0\}$ for $s = 1$, and $s-1, \dots, k-1$ otherwise. Return $(1, 0, i)$ if $i \geq j$, and $(0, D(s-1, i) + C(i+1, j), i)$ otherwise. Search, store predecessors, and backtrack the solution ending at k .

Lemma 7.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

Theorem 7.6 (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (8) and $1 \leq m \leq k$, both expected-participation and deterministic/pathwise frontiers through budget m , including*

an optimal codebook for every budget, require $O(mk)$ rational operations and comparisons and $O(mk + k)$ stored rational numbers and indices. The highest randomized level remains continuous. With total input length L and $S = O(k[L + \log(k+2)])$, a conservative Turing bound is $O(mkS^3)$. Sorting unordered distinct caps adds $O(k \log k)$ exact comparisons.

The proof is in Electronic Companion Section EC.1.13.

The retained divide-and-conquer implementation is a valid $O(mk \log(k+1))$ alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem. $O(k)$ exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

8 Promise Allocation Away from Saturation

The saturated promise is an endpoint of a larger allocation problem, not a normalization. Fix the branch caps and replace the root promise by $B \in [0, \bar{b}]$, where $\bar{b} = \sum_j \pi_j b_j$. Write $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for optimal value with at most m symbols under expected participation, pathwise participation allowing draw-specific intermediate tiers, and deterministic execution, respectively. The information interface remains unchanged. There is no codeword-dependent charge in this section, and terminal levels may range continuously over $[0, 1]$. The full-information value is $V_\infty(B)$.

8.1 An exact threshold for every root promise

Theorem 8.1 (Clipped service levels and exact alphabet). *Suppose f is strictly increasing and strictly concave, and each h_j is convex, nondecreasing, with $h_j(0) = 0$. For every $B \in [0, \bar{b}]$, full information uniquely implements*

$$Y_j = 0, \quad C_j = \min\{b_j, \tau\}, \quad \sum_j \pi_j \min\{b_j, \tau\} = B. \quad (21)$$

At $B = 0$ take $\tau = 0$, and at $B = \bar{b}$ take $\tau = \max_j b_j$. The minimum alphabet attaining this value under any of the three institutions is

$$n(B) = |\{\min(b_j, \tau) : 1 \leq j \leq k\}|.$$

In particular, for $0 \leq B \leq \min_j b_j$, one symbol attains $V_m^I(B) = f(B)$ for every $m \geq 1$ and $I \in \{E, P, D\}$.

Proof. For a full-information feasible policy, let $t_j = \mathbb{E}(Y_j + C_j) \leq b_j$. Replacing it on branch j by $Y_j = 0, C_j = t_j$ preserves the root promise and satisfies even pathwise participation. Jensen's inequality, strict increase of f , and $h_j \geq 0$ show that the replacement weakly improves value, strictly if $\mathbb{E}Y_j > 0$ or C_j is nonconstant. Hence it suffices to maximize $\sum_j \pi_j f(t_j)$ with $0 \leq t_j \leq b_j$ and $\sum_j \pi_j t_j = B$. Strict concavity implies uniqueness. If two uncapped coordinates differ, transferring equal probability-weighted amounts from the larger to the smaller strictly improves this sum; the same transfer shows that a capped coordinate below the common free level must be filled. Thus the unique optimizer is (21), with the endpoint cases interpreted as stated. Continuity of the clipped mean gives existence of τ .

Equality in these improvements requires deterministic terminal action $\min(b_j, \tau)$ on each branch. A symbol's terminal decoder cannot implement two different deterministic actions at the same public vertex. Its conditional output distribution is independent of the originating branch, including when fresh decoder randomization is permitted. Consequently distinct optimal actions require distinct symbols. Assigning one symbol per distinct action attains the lower bound under all institutions. When $B \leq \min b_j$, all optimal actions equal B . $\square$

For strictly ordered caps, the thresholds at which $n(B)$ changes are explicit: put $B_\ell = \sum_j \pi_j \min(b_j, b_\ell)$. The exact alphabet is $\ell + 1$ for $B_\ell < B \leq B_{\ell+1}$, $1 \leq \ell < k$, and it is one on $[0, B_1]$. At $B = B_\ell$ it is ℓ . Thus exact implementation has a promise-dependent phase diagram, rather than requiring k symbols at every promise.

Proposition 8.2 (Repeated caps at the saturated endpoint). *For $B = \bar{b}$, branches with a common cap b can be aggregated in every deterministic, expected, and pathwise frontier. Their aggregate probability is $p_b = \sum_{j:b_j=b} \pi_j$ and their aggregate intermediate cost is $h_b(y) = p_b^{-1} \sum_{j:b_j=b} \pi_j h_j(y)$. Quadratic costs therefore aggregate by their probability-weighted curvature. The exact alphabet threshold is the number of distinct caps. For arbitrary B , the full-information threshold remains (21), but aggregation of the restricted-alphabet problem with heterogeneous intermediate costs is not asserted.*

Proof. At saturation, the fixed-book expected rule chooses the same terminal mean and intermediate tier for all branches with the same cap. The deterministic and pathwise rules choose the same highest eligible terminal level. Summing their rewards gives precisely the aggregate cost. Optimize over the common books and apply Theorem 8.1 for the threshold. Away from saturation the branch promises themselves can differ, so the saturated aggregation argument does not apply. $\square$

8.2 Uniform transport between any two root promises

Assume in addition that f is Lipschitz with constant L_f on $[0, 1]$ and each h_j is Lipschitz with constant H_j on $[0, b_j]$; put $H = \max_j H_j$. These constants are independent of the alphabet budget. For quadratic primitives one may take $L_f = r$ and $H = \max_j \gamma_j b_j$. The smaller intermediate domain is sufficient because the mean and pathwise replacements used below have tiers at most b_j .

Theorem 8.3 (All-promise value and institutional-gap stability). *For any $0 \leq B_1 \leq B_2 \leq \bar{b}$, $\Delta = B_2 - B_1$, every $m \geq 1$, and $I \in \{E, P, D\}$,*

$$V_m^I(B_2) - L_f \Delta \leq V_m^I(B_1) \leq V_m^I(B_2) + H \Delta. \quad (22)$$

If $G_m(B) = V_m^E(B) - V_m^P(B)$, then

$$|G_m(B_2) - G_m(B_1)| \leq (L_f + H) \Delta. \quad (23)$$

The same bound holds for changes in the loss $V_\infty(B) - V_m^I(B)$. In particular, a certified strict advantage $G_m(B_0) > 0$ persists for every feasible B with $|B - B_0| < G_m(B_0)/(L_f + H)$.

Proof. First contract any policy at $B_2 > 0$ by multiplying all intermediate and terminal actions by $\alpha = B_1/B_2$. The decoder uses the same symbols, and both participation conventions remain feasible. Intermediate cost weakly decreases. The decrease in terminal reward is at most $L_f(1 - \alpha)\mathbb{E}C \leq L_f\Delta$, where expectations include branch weights. The case $B_2 = 0$ is immediate. Taking suprema gives the left inequality.

For the other direction, start from a policy at B_1 . Under expected participation, replace the terminal decoder given a symbol by its mean and the intermediate tier on each branch by its mean. These changes weakly improve reward, preserve the alphabet, and preserve expected obligations. Write $\mu_j = \mathbb{E}C_j$, $y_j = \mathbb{E}Y_j$, and $s_j = b_j - y_j - \mu_j \geq 0$. If $B_1 < \bar{b}$, put $\theta = \Delta/(\bar{b} - B_1)$ and replace y_j by $y_j + \theta s_j$. The total added obligation is $\theta \sum_j \pi_j s_j = \Delta$. Its cost is at most $H\Delta$, and the book and its encoding probabilities are unchanged.

For pathwise participation, replace fresh terminal decoder randomness conditional on symbol z by its mean c_z . Conditional independence from the earlier tier ensures $Y_j + c_z \leq b_j$ almost surely after this replacement; terminal concavity weakly improves reward. Use the draw-specific tier

$$Y'_j = Y_j + \theta(b_j - Y_j - c_z).$$

This is admissible at the stipulated intermediate timing, remains between Y_j and $b_j - c_z$, and adds expected obligation Δ with cost at most $H\Delta$. It requires no extra information at the terminal gateway. The deterministic case is the same construction without random draws. If $B_1 = \bar{b}$, necessarily $\Delta = 0$. Taking suprema, or limits of near-optimal policies, proves the right inequality without requiring a maximizer.

For each institution the value increment lies in $[-H\Delta, L_f\Delta]$. Subtracting two such increments proves the institutional-gap bound. The same constructions apply with unrestricted full information, yielding the loss bound. Every transformation preserves the original number of writable symbols; no concavity of the optimized codebook envelope is assumed. $\square$

In the equal-probability three-branch example of Proposition 5.2, $\bar{b} = 1/2$, $G_2(\bar{b}) = 11/96$, and $L_f + H = 2 + 3/4 = 11/4$. Therefore

$$G_2(B) \geq \frac{11}{96} - \frac{11}{4} \left(\frac{1}{2} - B \right) > 0 \quad \text{for } \frac{11}{24} < B \leq \frac{1}{2}. \quad (24)$$

This is a sufficient certified interval of strict advantage, not a maximal interval or an exact nonsaturated loss formula. The global cell algorithm below computes exact values outside this conservative interval as well. At the other endpoint, $B \leq 1/4$ gives zero loss with one symbol. In between, cap anchoring at the original b_j and pathwise-deterministic equality do not follow from the saturated proofs. Indeed the one-symbol full optimum $C = B < \min b_j$ already shows why original-cap anchoring cannot hold for all promises. The decision insight persists without implying that the saturated algorithm solves every interior promise.

8.3 The nonsaturated allocation problem

The larger problem has an exact reduction even when it does not have the saturated finite-anchor reduction. For a fixed book $c = (c_1, \dots, c_s)$ let F_c be its concave interpolant and define

$$v_{j,c}(t) = F_c(\min\{t, c_s\}) - h_j((t - c_s)_+), \quad c_1 \leq t \leq b_j. \quad (25)$$

Proposition 8.4 (Joint promise allocation and codebook design). *The expected-participation value at any B is exactly*

$$V_m^E(B) = \max_{\substack{|c| \leq m, 0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq t_j \leq b_j, \sum_j \pi_j t_j = B}} \sum_j \pi_j v_{j,c}(t_j). \quad (26)$$

For a fixed book this is a separable concave resource-allocation problem. Branch targets are the chosen t_j , not mechanically the original caps. Saturation fixes $t_j = b_j$ and recovers the earlier frontier.

Proof. Set $t_j = \mathbb{E}(Y_j + C_j)$ for an arbitrary feasible policy. The mean replacements in Theorem 5.1 are still valid. For a given t_j , maximizing $F_c(\mu) - h_j(t_j - \mu)$ over $c_1 \leq \mu \leq \min(t_j, c_s)$ selects its largest feasible mean by monotonicity. Conversely that mean and its adjacent lottery implement (25) with intermediate tier $(t_j - c_s)_+$. This proves both directions of (26). The response is concave: to the left of c_s it is F_c , to the right it is $f(c_s) - h_j(t - c_s)$, and the one-sided slopes drop from a nonnegative chord slope to a nonpositive cost slope. This also covers a singleton book. Allocation with a common supporting multiplier, allowing subgradient intervals at kinks, solves the fixed-book problem. No interchange of codebook optimization and a dual minimization is used. $\square$

Electronic Companion Proposition EC.6.1 gives an exact supporting-price algorithm and a separately checkable certificate for every fixed-book allocation. Exhaustive certified catalog search extends that computation to all promises on small catalogs; its exponential cost is stated explicitly.

Theorem 9.1 extends the fixed-book response to every realization ceiling. Theorems 9.3 and 11.1 solve the continuous outer design and certify its catalog approximation; thus the inner allocation is no longer the endpoint of the nonsaturated analysis.

9 Joint Continuous Design at Every Promise

We now optimize the continuous alphabet, branch promises, bounded realization exposure, and actual selected-level charges in one problem. Fix $B \in [0, \bar{b}]$ and $\delta \geq 0$. On branch j , an intermediate tier y_j is committed before a terminal lottery is drawn, and

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (27)$$

A book c has at most m distinct levels in $[0, 1]$ and incurs $\sum_{z \in c} \rho(z)$, once per selected level. Let $\mathcal{V}_m^\delta(B; \rho)$ denote maximum operating payoff less this charge. A finite catalog restriction is denoted $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho)$. The expectation cap is not replaced by the weaker realization ceiling. Taking $\delta \geq 1$ makes the realization

restriction redundant at the canonical optimum below. Taking $\delta = 0$ gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

9.1 Eligibility prefixes replace saturated targets

For a strictly ordered book $c = (c_1, \dots, c_s)$ and branch j , define

$$c^{j,\delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j,\delta},$$

and let $F_{j,c}^\delta$ be the piecewise-affine interpolant of f on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

Theorem 9.1 (Canonical allocation under a realization ceiling). *Suppose f is strictly increasing and concave and h_j is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. Its optimal branch response at target $c_1 \leq t \leq b_j$ is*

$$v_{j,c}^\delta(t) = F_{j,c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (28)$$

It is attained by $y = (t - d_j)_+$ and an adjacent lottery of mean $\min(t, d_j)$ in the eligible prefix. Consequently

$$\mathcal{V}_m^\delta(B; \rho) = \max_{\substack{|c| \leq m, \ c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j,c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (29)$$

The fixed-book inner problem is concave. With zero charges, $\mathcal{V}_m^0(B; 0) = V_m^P(B)$ and $\mathcal{V}_m^\delta(B; 0) = V_m^E(B)$ for $\delta \geq 1$, at every feasible B .

Proof. Every used terminal level satisfies $C \leq b_j + \delta$ because $y \geq 0$. Writing $\mu = \mathbb{E}C$, the payoff is at most $F_{j,c}^\delta(\mu) - h_j(t - \mu)$. Both terms are nondecreasing as μ increases on $[c_1, \min(t, d_j)]$, so this upper bound is maximized at its right endpoint. If $t \leq d_j$, the adjacent lottery has $y = 0$ and all its support lies below $b_j + \delta$. If $t > d_j$, it is the singleton d_j and $y + d_j = t \leq b_j$. Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of $-h_j(t - d_j)$. The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval $[c_1, \bar{b}]$.

For pathwise participation with possibly draw-specific Y , all used C satisfy $C \leq b_j$. At fixed $t = \mathbb{E}(Y + C)$, concavity of f on the discrete support and Jensen's inequality for h_j give the same upper bound with $\delta = 0$ and $\mathbb{E}Y = t - \mu$. The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the same mean argument has no eligibility truncation; $\delta \geq 1$ includes every level in $[0, 1]$. These observations establish the two endpoint identities and both directions of (29). $\square$

For quadratic primitives (8), chord $[u, v]$ has slope $r - q(u + v)/2$. On branch j its target capacity is

$$\left[\min\{b_j, v\} - u \right]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at $t_j = c_1$, fill these positive slopes in their common decreasing order, followed by zero-cost intermediate capacity and the quadratic tails. A water-level sweep over events $\gamma_j(b_j - d_j)_+$ fills the remaining targets. It uses $O(ks + k \log(k + 1))$ rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (30)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

9.2 A finite exact reduction of the continuous outer problem

Fix $s \leq m$ and temporarily allow $0 \leq c_1 \leq \dots \leq c_s \leq 1$. For each branch select one of $2s - 1$ modes. A singleton mode S_ℓ and an adjacent-lottery mode L_ℓ have the following closed descriptions:

$$S_\ell : \quad c_\ell \leq t_j \leq b_j, \quad g_j^S = rc_\ell - \frac{q}{2}c_\ell^2 - \frac{\gamma_j}{2}(t_j - c_\ell)^2; \quad (31)$$

$$L_\ell : \quad c_\ell \leq t_j \leq c_{\ell+1}, \quad t_j \leq b_j, \quad c_{\ell+1} \leq b_j + \delta, \quad g_j^L = rt_j - \frac{q}{2}[(c_\ell + c_{\ell+1})t_j - c_\ell c_{\ell+1}]. \quad (32)$$

For S_ℓ , use $y_j = t_j - c_\ell$ and terminal level c_ℓ . For L_ℓ , use $y_j = 0$ and the adjacent unbiased lottery. At $c_\ell = c_{\ell+1}$ the constraints force $t_j = c_\ell$, and the formula is the singleton reward. Therefore collisions are not discarded as singular probability formulas. The quadratic interpolation identity eliminates the ratio in the lottery probabilities before codewords become optimization variables.

Each joint mode assignment, the ordered-book inequalities, and $\sum_j \pi_j t_j = B$ define a compact polytope. If $\rho(c) = \alpha c^2 + \beta c + \eta$ is nonnegative on $[0, 1]$, its objective

$$Q(c, t) = \sum_j \pi_j g_j(c, t) - \sum_{\ell=1}^s \rho(c_\ell) \quad (33)$$

is a quadratic polynomial, not necessarily concave. Nonnegative charges, rather than signs of their coefficients, are the assumption. For example, a decreasing nonnegative certification charge is allowed.

Lemma 9.2 (Finite stationary-face search). *A quadratic function on a nonempty compact polytope attains its maximum at a point that is either a vertex or the unique stationary point on the affine hull of some face with negative-definite restricted Hessian. With rational input data, some maximizing point and its value are rational and can be found by finitely many rational linear solves and feasibility tests.*

Proof. Among all faces whose relative interiors contain a global maximizer, choose one of least dimension and a maximizing point x in its relative interior. The derivative along the face's tangent space vanishes and

the restricted Hessian is negative semidefinite. If this restriction has a nonzero null direction z , the quadratic is constant along $x + az$ in the affine hull: both its linear and quadratic terms in a vanish. Compactness makes this line reach a proper face while remaining feasible, contradicting the choice of dimension. The restriction is therefore negative definite unless the tangent space is zero dimensional.

Enumerate independent subsets of the defining inequalities, together with the equality constraints, and solve the stationarity equations on each associated affine space. On the face just identified the bordered stationarity matrix is nonsingular; at a vertex, the tangent space is empty and the same conclusion holds. Test feasibility in every original inequality and retain the largest objective among all feasible candidates. Each retained candidate is feasible, and at least one is a global maximizer. Rational elimination gives the rationality claim. Singular bordered systems may be skipped because the argument guarantees a nonsingular maximizing face of smaller dimension; it does not assume nondegeneracy of the original polytope. $\square$

Theorem 9.3 (Exact joint continuous design). *For rational quadratic primitives, rational caps and probabilities, rational $B \in [0, \bar{b}]$ and $\delta \geq 0$, and a rational quadratic charge nonnegative on $[0, 1]$, the maximum of (33) over the closed cells (31)–(32), for every $1 \leq s \leq m$, equals $\mathcal{V}_m^\delta(B; \rho)$. The optimum is attained by a rational book, rational targets, and rational lottery probabilities. Deleting the lottery modes gives the exact deterministic frontier at every promise, with the same charges.*

An exact algorithm enumerates at most $(2s - 1)^k$ cells at size s . After eliminating one target, each cell has dimension at most $s + k - 1$ and at most $s + 1 + 4k$ inequalities. Enumerating its independent active sets and applying Lemma 9.2 gives a finite global algorithm with at most

$$\sum_{s=1}^m (2s - 1)^k 2^{s+1+4k} \quad (34)$$

linear systems, each of polynomial dimension and rational encoding length in the input size. The sharper rank-limited count replaces 2^{s+1+4k} by $\sum_{r=0}^{\min\{s+k-1, s+1+4k\}} \binom{s+1+4k}{r}$. A lottery branch may impose four separate inequalities: its two chord bounds, its target cap, and its realization eligibility bound. Neither of the latter two is redundant at a general positive tolerance. This is an exponential reference algorithm, not a polynomial-time claim or an extension of the saturated Monge complexity.

Proof. Theorem 9.1 turns every optimal book and target allocation into singleton or adjacent-lottery branch modes represented in these cells. Conversely every cell point implements a feasible policy by the displayed reconstruction; all constraints, including the exact root promise, hold. If ordered slots collide or some slots are unused, deleting redundant slots cannot increase a nonnegative charge. The resulting smaller book and its canonical branch rules appear among the enumerated sizes. Hence overcounting charges at duplicate slots cannot change the maximum across all sizes. These two directions prove equality with the original optimization problem, not merely a relaxation or a fixed-book upper bound.

There are finitely many compact cells and each has a continuous quadratic objective. Lemma 9.2 supplies a rational maximizing candidate on each nonempty cell. Adjacent distinct rational levels reconstruct rational probabilities; coincident levels reconstruct a singleton. The mode count, inequality count, and all-subsets upper bound give (34). Standard determinant bounds for rational linear systems give polynomial encoding

length for each candidate. Singleton-only modes exactly represent deterministic terminal assignments and intermediate tiers, proving the deterministic statement. $\square$

The reduction solves the unrestricted continuous outer problem at an interior promise, rather than relabeling fixed-book allocation as global design. Its contract-specific content is the eligibility-prefix reduction and the closed quadratic elimination of lottery probabilities with no loss at collisions. Rational quadratic optimization and active-set reasoning are classical; polynomial-size witnesses are not a new general complexity principle (Del Pia et al., 2017). The fast saturated Monge theorem remains valuable precisely because the general finite search is exponential.

9.3 Pathwise comparison, repeated caps, and small charges

Theorem 9.3 computes $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for every rational promise, using respectively all lottery modes at $\delta = 1$, all modes at $\delta = 0$, and singleton modes. No claim of pathwise-deterministic equality away from saturation is needed for this characterization. Section EC.11 compares their exact values along an entire declared parameter path. The zero-memory-loss interval $B \leq \min b_j$ in Theorem 8.1 remains important: a positive institution gap is not expected throughout the feasible promise range.

Proposition 9.4 (Pathwise randomization can matter in the interior). *For one branch with $b = 9/10$, $B = 9/20$, $r = 2$, $q = \gamma = 1$, two available symbols, and the nonnegative charge $\rho(c) = 3c(1 - c)$,*

$$\mathcal{V}_2^0(B; \rho) = \frac{171}{400} > \mathcal{V}_2^D(B; \rho) = \frac{9}{160}.$$

The pathwise optimum uses book $(0, 9/10)$, equal draw probabilities, and no intermediate tier. The deterministic optimum uses the single level $9/20$.

Proof. A deterministic policy uses only one level $0 \leq c \leq B$; deleting unused levels weakly lowers its charge. Its net payoff is $-81/800 - (11/20)c + 2c^2$, whose maximum over $[0, B]$ is $9/160$ at B . Every nonsingleton canonical pathwise mode has $0 \leq u \leq B \leq v \leq b$, and net payoff $2B - (u+v)B/2 + uv/2 - 3u(1-u) - 3v(1-v)$. This is convex in (u, v) , so its maximum on that rectangle is at a corner. Checking its four corners, together with the singleton case, gives $(u, v) = (0, 9/10)$ and value $171/400$. Both realizations meet the hard cap and their mean is exactly B . $\square$

This example uses a specified nonnegative, nonmonotone level charge; it is not a counterexample asserted for the uncharged problem. It shows why the saturated pathwise collapse cannot simply be imported into the unified interior design model.

Proposition 9.5 (Homogeneous repeated-cap aggregation). *For the randomized institutions in (27), branches with identical caps and identical intermediate-cost functions can be aggregated by adding their probabilities, at every B , δ , and codeword charge. This holds for each fixed book and after global book optimization.*

Proof. Such branches have identical eligible prefixes and the same concave response (28). Replacing their targets by their probability-weighted mean preserves the root promise and does not lower total reward by

Jensen's inequality. Replicating an aggregate target gives the converse inequality. Charges depend only on the common book. This argument does not aggregate heterogeneous costs or establish a corresponding deterministic restriction. $\square$

The strict off-cap conclusion is also stable under genuinely selected-level charges. In Proposition 6.3, the uncharged two-level book $(1/4, 5/8)$ beats every book of at most two cap levels by $1/1280$. If $0 \leq \rho(c) \leq \eta$ on $[0, 1]$ and $\eta < 1/2560$, evaluating this same off-cap book incurs less than that gap, while a cap-only book cannot improve its uncharged payoff after a nonnegative charge. Thus no charged optimum is cap-only. This statement preserves the off-cap property, not necessarily the numerical location $5/8$.

9.4 What the shaped-charge example does and does not imply

Proposition 9.4 establishes a logical separation for a specified nonmonotone charge. It does not establish that pathwise randomization typically saves engineering or certification cost. The following result gives a sharp institutional boundary for that example rather than assigning its specially shaped charge an unsupported operational interpretation.

Proposition 9.6 (One branch with a monotone charge). *For continuously chosen levels and a single branch, concave terminal reward, convex intermediate cost, and a nonnegative nondecreasing charge per selected level, a deterministic one-level policy weakly dominates every randomized policy at each feasible promise. This holds for every alphabet budget and every realization tolerance, including the full pathwise institution with draw-specific intermediate service.*

Proof. For an arbitrary feasible policy write $\mu = \mathbb{E}C$ and $\bar{y} = \mathbb{E}Y = B - \mu$. The deterministic level μ with intermediate service $\bar{y}$ attains total service $B \leq b$, and Jensen's inequalities improve both operating terms. The old book contains a largest used level $c_{\max} \geq \mu$. Nonnegativity and monotonicity imply that its total charge is at least $\rho(c_{\max}) \geq \rho(\mu)$. Thus the replacement is feasible, uses one level, and does not lower net payoff. $\square$

Consequently the one-branch separation in Proposition 9.4 necessarily uses a charge outside this monotone class. The result covers zero charges, nondecreasing affine or convex charges, and a fixed opening fee plus a nondecreasing surcharge. It does not claim deterministic collapse for arbitrary heterogeneous multi-branch instances; the common alphabet couples their means. The original example and its proof remain intact as a mathematical possibility, not evidence of a generic operational advantage.

10 Eligibility-Prefix Decomposition

The closed-cell formulation establishes exact continuous attainability. We now exploit the ordered interface to avoid enumerating either branch modes or catalog subsets in a price subproblem. The resulting bounds support finite global search and remain valid when that search stops early. This distinction matters: a price relaxation is not, without an additional equality, the original optimum.

10.1 A reusable ordered-interface allocation class

Let $\mathcal{A} = (a_1 < \dots < a_N) \subset [0, 1]$ be a catalog with nonnegative opening charges ρ_i . The common reward f is strictly increasing and concave. Branch j has weight π_j , target cap b_j , nondecreasing convex shortfall cost h_j with $h_j(0) = 0$, and an eligibility threshold $\tau_j \geq b_j$. Its response is (28) with eligible levels $c \cap [0, \tau_j]$. The renewal model has $\tau_j = b_j + \delta$; the theorem below permits distinct thresholds without requiring a common overrun tolerance. This abstract class needs no acceptance or renewal interpretation. It describes any ordered shared interface with a common concave output reward, separable shortfall costs, and these target and support restrictions.

Write $a_* = \min\{B, \min_j b_j\}$. Assume $a_1 \leq a_*$, since otherwise no feasible book exists, and replace m by $\min\{m, N\}$. Restrict books to $c_1 \leq a_*$; this retains exactly those books that can meet the root promise. At a real price λ , define

$$g_\lambda(x) = f(x) - \lambda x, \quad \phi_j(z; \lambda) = \max_{0 \leq y \leq z} \{-h_j(y) - \lambda y\}, \quad (35)$$

$$D(\lambda) = \lambda B + \max_{\emptyset \neq c \subset \mathcal{A}, |c| \leq m, c_1 \leq a_*} \left[\sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{a_i \in c} \rho_i \right]. \quad (36)$$

Weak duality gives $\mathcal{V}_{m,\mathcal{A}}^\delta(B; \rho) \leq D(\lambda)$ in the renewal specialization. For quadratic shortfall costs, (35) is evaluated at $y = 0$ if $\lambda \geq 0$, and at $y = \min\{z, -\lambda/\gamma_j\}$ otherwise; when $\gamma_j = 0$ and $\lambda < 0$, use $y = z$.

Call an arc $u < v$ admissible when $g_\lambda(v) \geq g_\lambda(u)$. Define its reward and the reward for stopping at u by

$$E_\lambda(u, v) = \sum_{j: u \leq b_j < v} \pi_j \begin{cases} g_\lambda(u) + \frac{b_j - u}{v - u} [g_\lambda(v) - g_\lambda(u)], & v \leq \tau_j, \\ g_\lambda(u) + \phi_j(b_j - u; \lambda), & v > \tau_j, \end{cases} \quad (37)$$

$$T_\lambda(u) = \sum_{j: b_j \geq u} \pi_j [g_\lambda(u) + \phi_j(b_j - u; \lambda)]. \quad (38)$$

An edge accounts only for branches whose target caps it crosses. It therefore carries a contractual allocation value, not an ordinary nearest-neighbor quantization error.

Theorem 10.1 (Polynomial price decomposition). *For every λ , some maximizer in (36) has nondecreasing g_λ along its ordered book. Its priced operating value is the sum of (37) over consecutive levels plus (38) at the last level. Set*

$$W_1(i) = T_\lambda(a_i), \quad (39)$$

$$W_r(i) = \max \left\{ T_\lambda(a_i), \max_{\substack{\ell > i \\ g_\lambda(a_\ell) \geq g_\lambda(a_i)}} [E_\lambda(a_i, a_\ell) - \rho_\ell + W_{r-1}(\ell)] \right\}.$$

Then $D(\lambda) = \lambda B + \max_{a_i \leq a_*} [W_m(i) - \rho_i]$. With constant-time evaluation of ϕ_j , computing a maximizing book and this bound takes $O((k+m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, allowing both k and m to vary. Rational quadratic data and a rational price give an exact rational algorithm with polynomial

bit complexity. General costs use at most $O(kN^2)$ calls to the scalar oracle (35) instead.

Proof. If $\lambda < 0$, strict increase of f makes g_λ strictly increasing. If $\lambda \geq 0$, intermediate service cannot improve priced payoff, and a branch maximizes the interpolant of g_λ on its eligible prefix, truncated at its target cap. Concavity implies that the values along any book increase up to their first maximum and never subsequently increase. All levels after that first maximum may be deleted: for a branch below the maximizing level, either that level is ineligible and all later levels are also ineligible, or later chords cannot improve the concave interpolant before the cap; for a branch above it, no later value beats the maximum. Nonnegative charges make deletion weakly beneficial. The retained book has only admissible arcs.

Consider $b_j \in [u, v)$ on such a book. If $v \leq \tau_j$, the best priced target is b_j on the chord $[u, v]$. For $\lambda \geq 0$ its slope is nonnegative and a singleton tail contributes no positive adjustment. For $\lambda < 0$, a singleton tail contributes at most $-\lambda(b_j - u)$, whereas the chord contributes this amount plus a strictly positive reward increment when $b_j > u$. If $v > \tau_j$, every later level is ineligible; the optimum is a singleton at u followed by the best shortfall in (35). Branches beyond the final level have the same singleton-tail form. These disjoint cap intervals cover all branches because the first level is at most $\min_j b_j$. This proves the decomposition, including equality cases.

The acyclic recurrence chooses either to stop or to append the next admissible level, paying each selected charge once. It enumerates paths implicitly rather than enumerating books. Computing edges costs $O(kN^2)$; the dynamic-programming layers cost $O(mN^2)$. For rational quadratic data, every scalar oracle value uses a bounded number of rational operations, and each path value sums at most $O(km)$ such terms. Numerator and denominator lengths are polynomial in the input bit length; comparisons and path reconstruction therefore preserve polynomial bit complexity. $\square$

Pricing and constrained shortest-path bounds are established optimization mechanisms (Handler and Zang, 1980). The model-specific result here is that one price makes an otherwise nonseparable shared-book allocation decompose into these cap-crossing edges, despite heterogeneous shortfall costs, branch-specific eligibility, and nonsaturated targets. The theorem does not assert a polynomial algorithm for the unrelaxed joint problem or exchange its maximization with minimization over prices.

10.2 Prefix bounds and checkable global search

A node is a mandatory initial segment $p = (i_1, \dots, i_s)$ of a book; its descendants may append only larger indices. If the priced values along p never decrease, its exact priced completion value is

$$U_p(\lambda) = \lambda B + \sum_{h=1}^{s-1} E_\lambda(a_{i_h}, a_{i_{h+1}}) - \sum_{h=1}^s \rho_{i_h} + W_{m-s+1}(i_s). \quad (40)$$

If a decrease occurs in p , concavity places its first priced maximum within p and every larger new level is useless to the priced allocation. In that case, use the fixed-prefix branch-support value, with every mandatory charge retained. This is again the exact priced maximum over completions; additional nonnegative charges cannot help. The empty prefix uses (36).

Theorem 10.2 (Finite search and an anytime global interval). *Fix any nonempty finite set of prices. At each prefix use the smallest corresponding $U_p(\lambda)$, evaluate feasible books by the exact inner allocator, and retain the largest feasible net value L . Partition an unpruned node into its own book, when nonempty, and all one-index extensions. Prune a node only when its upper bound is at most L . This procedure terminates after finitely many prefixes and returns the exact catalog optimum. At any interruption it returns*

$$L \leq \mathcal{V}_{m,\mathcal{A}}^\delta(B;\rho) \leq U = \max\{L, \max_{p \text{ unresolved}} \min_{\lambda} U_p(\lambda)\}. \quad (41)$$

For a uniform mesh of spacing h , Theorem 11.1 further yields the continuous interval $[L, U + K_m h]$. The optimization gap $U - L$ and discretization error $K_m h$ are separate and observable.

Proof. The preceding completion argument proves each price bound, so their minimum is valid by weak duality. Every feasible book either equals a nonempty node or has a unique next index. Hence the prescribed children and the node’s own book are a disjoint exhaustive partition; book size is at most m . A pruned subtree cannot improve the incumbent. The maximum of unresolved upper bounds and the incumbent therefore bounds every feasible book at every stage. If all subtrees have been resolved, the two bounds coincide. The mesh transport is a separate one-sided inequality between the continuous optimum and the exact catalog optimum, so substituting the catalog upper bound proves the last claim. $\square$

A global witness consists of the feasible winning lottery, every split and its complete child list, upper bounds and prices at pruned or unresolved nodes, and a fixed-book supporting price for each explicitly evaluated node book. A checker can rebuild price tables and verify the partition without replaying the search order or trusting the incumbent-selection code. This is a stronger object than an inner multiplier or a source hash. Equality of one root price bound with a feasible payoff already certifies global catalog optimality without any branching. Worst-case search is still exponential in the alphabet budget; neither this result nor the mesh transfer is an FPTAS claim.

Additional decomposition contribution. Theorems 10.1–10.2 use classical Lagrangian bounds and an acyclic path recurrence. Their structural increment is the exact eligibility-aware cap-crossing reward and prefix-completion dominance for nonsaturated allocation with heterogeneous shortfall costs. The generic search framework is not claimed as new. Proposition 11.3 establishes sharpness for uniform catalogs, not a lower bound for adaptive grids. The continuous stationary-face mechanism remains generic; its contractual closed-cell reduction remains the structural contribution.

11 Catalog Approximation Without Promise or Risk Slack

Finite catalogs have two different interpretations. An externally certified menu is a restriction of the feasible technology, for which an exact catalog optimum is the relevant answer. Alternatively the operator may choose a numerical catalog to approximate continuous design. The latter interpretation needs a quantitative comparison to the unrestricted optimum; exactness on the catalog alone does not supply it.

Let $\mathcal{A} \subset [0, 1]$ contain zero and one, and let h be its largest consecutive spacing. Assume f is increasing, concave, and L_f -Lipschitz, and each h_j is convex, nondecreasing, and H_j -Lipschitz on $[0, b_j]$. Let the nonnegative charge ρ be L_ρ -Lipschitz on $[0, 1]$. Define

$$K_m = L_f + \sum_j \pi_j H_j + m L_\rho. \quad (42)$$

For the quadratic model, one can take $L_f = r$ and $H_j = \gamma_j b_j$. Neither the root promise nor the risk tolerance is discretized.

Theorem 11.1 (Uniform exactly feasible catalog certificate). *For every $B \in [0, \bar{b}]$, every $\delta \geq 0$, and every $m \geq 1$,*

$$0 \leq \mathcal{V}_m^\delta(B; \rho) - \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq K_m h. \quad (43)$$

More strongly, every canonical continuous-book policy has a catalog policy with no more symbols that preserves each branch target exactly, satisfies the same realization ceiling exactly, and loses at most $K_m h$ in net payoff. The conclusion includes the pathwise endpoint $\delta = 0$ and requires no strict feasibility margin.

Proof. Round each original level c downward to $a(c) = \max\{a \in \mathcal{A} : a \leq c\}$ and merge collisions. Fix a branch target t and let d be its old eligible top. Every old eligible level remains eligible after rounding. In particular the new eligible top d' satisfies $d' \geq a(d) \geq d - h$. The old canonical terminal mean is $\mu = \min(t, d)$; coupling its draws to their rounded levels gives a feasible-support lottery of mean $\tilde{\mu} \in [\mu - h, \mu]$ and expected terminal reward at least the old reward minus $L_f h$.

Do not retain those coupled probabilities. Recompute the canonical lottery on the rounded eligible prefix at the *same* target t . Its terminal mean $\mu' = \min(t, d')$ is at least $\tilde{\mu}$: every rounded old draw is at most d' and its mean is at most t . The new interpolant is increasing, and at mean $\tilde{\mu}$ it dominates the coupled lottery reward. Therefore its reward at μ' also dominates that reward. Moreover $\mu' \geq \mu - h$, so its intermediate tier $t - \mu'$ exceeds the old tier by at most h . The branch operating-payoff loss is at most $(L_f + H_j)h$. Theorem 9.1 ensures exact mean and risk feasibility for this reoptimized lottery, including when the risk bound is binding. All t_j are unchanged, so the root promise remains exactly B .

For each original level, $\rho(a(c)) \leq \rho(c) + L_\rho h$. After merging, the total charge is no larger than the sum of these rounded-slot charges because charges are nonnegative. At most m slots were present. Summing the branch bounds and charge bound gives $K_m h$. Apply the construction to an optimizer, or to a sequence approaching the supremum, and use inclusion of catalog policies in continuous policies for the other inequality. $\square$

One-sided rounding is necessary but not sufficient without the probability repair. For example, let $b = t = 1/2$, $\delta = 1/8$, and old support $(1/4, 5/8)$ with upper probability $2/3$. Rounding the lower level to zero while retaining the probabilities reduces the mean to $5/12$. Compensating by $y = 1/12$ produces a maximum obligation $17/24 > 5/8$, violating the risk limit. Instead, the new canonical lottery on $(0, 5/8)$ uses upper probability $4/5$ and $y = 0$. It preserves the target and the ceiling exactly. This is the distinction between unconstrained quantizer rounding and the contractual transport in Theorem 11.1.

Corollary 11.2 (Certified approximation for a fixed alphabet budget). *For rational quadratic operating primitives and a nonnegative charge whose rational catalog evaluations and Lipschitz bound are supplied, enumerate every feasible catalog book of size at most m and solve its allocation by (30). With $N = |\mathcal{A}|$, this gives an exact catalog optimum in*

$$O\left(\left[\sum_{s=1}^{\min(m,N)} \binom{N}{s}\right] (km + k \log(k+1))\right) \quad (44)$$

rational arithmetic/evaluation operations. A uniform mesh with denominator $\lceil K_m/\varepsilon \rceil$ (or one when that ceiling is zero) gives additive error at most ε . The algorithm is polynomial in k , K_m/ε , and the bit complexity of the sampled data for fixed m ; it is not claimed to be polynomial in a variable m , nor is this a multiplicative approximation ratio.

Proof. Increasing subsets exhaust the allowed books and the supporting-price algorithm optimizes every inner problem exactly. Take the largest charge-adjusted value and apply Theorem 11.1. The mesh includes zero, so every feasible promise is implementable even when it is smaller than the first positive grid point. The bound counts evaluation work; a black-box charge oracle is not assumed to have unit bit complexity unless that is explicitly part of the input model. $\square$

The guarantee is uniform in the two contractual constraints. For any institution pair, subtracting their certified value intervals gives a certified interval for the institutional gap. This permits a global accuracy statement away from saturation without presuming that original cap anchors or Monge order survive. When $B = \bar{b}$, the faster charged path recurrence in Section 12 replaces subset enumeration. When the saturated expected problem is uncharged, the continuous Monge algorithm avoids a mesh entirely. Thus the specialized algorithms are faster members of one feasible-policy hierarchy, while (43) connects numerical catalogs to continuous risk-and-charge design.

11.1 The uniform-mesh rate is sharp

The linear bound in Theorem 11.1 reflects exact feasibility, not only an insensitive Lipschitz estimate. It cannot be uniformly replaced by a higher-order rate for the declared uniform catalogs.

Proposition 11.3 (A fixed-input lower bound). *Take one branch and one symbol, $b = B = 1/2$, $f(c) = c - c^2/2$, $h(y) = y^2/2$, and zero charges. For every risk tolerance and every odd integer $N \geq 3$, the uniform catalog with spacing $h = 1/N$ has loss*

$$\mathcal{V}_1^\delta(B; 0) - \mathcal{V}_{1, \mathcal{A}_N}^\delta(B; 0) = \frac{h}{4} + \frac{h^2}{4}.$$

Thus the worst-case $O(h)$ rate is order-sharp, even for a single fixed input with no opening charges and no randomization.

Proof. With one level, $c \leq B$ and $y = B - c$. Payoff $f(c) - (B - c)^2/2$ is strictly increasing on $[0, B]$. Its continuous maximizer is B , with value $3/8$. For odd N , the largest feasible catalog point is $B - h/2$.

Substitution gives the displayed loss. All realizations equal total service B , so risk tolerance plays no role. The sequence of odd N uses the same primitives throughout. $\square$

This is a lower bound for uniform catalogs, not for all adaptive discretizations. Adding the cap $1/2$ removes this particular error. It also does not rule out faster convergence on other fixed instances. The search interval in (41) is therefore useful even when its optimization gap has already vanished: refinement addresses a genuinely different source of error.

12 Joint Design of a Risk-Limited Certified Alphabet

A finite certification catalog gives a general intermediate institution and a nontrivial codebook-dependent implementation problem. Let $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$ be the allowable terminal levels, with $a_1 \leq b_1$. Selecting a level a incurs an arbitrary nonnegative charge $\rho(a)$; for example, it may combine a declared precision/storage charge and a per-level certification charge in the same units as service loss. These are stipulated prices, not estimated operational costs. A book is any subset of $\mathcal{A}$ of size at most m . The objective includes the charges of its actual selected levels, not merely its declared capacity.

The root promise is saturated in this section. Intermediate service y_j is chosen before the symbol draw, with

$$y_j + \mathbb{E}C_j = b_j, \quad y_j + C_j \leq b_j + \delta \quad \text{almost surely.} \quad (45)$$

This explicitly defined institution has the two familiar endpoints: $\delta = 0$ attains the pathwise/deterministic catalog optimum, while $\delta \geq 1$ removes the overrun restriction. The pre-draw restriction is not asserted without loss for arbitrary intermediate risk protocols.

For adjacent selected levels $u < v$, assign to this edge the branches $u \leq b_j < v$ and define

$$K_\delta(u, v) = \sum_{j: u \leq b_j < v} \pi_j \ell_j^\delta(u, v), \quad (46)$$

where, on that interval,

$$\ell_j^\delta(u, v) = \begin{cases} f(b_j) - \frac{v - b_j}{v - u} f(u) - \frac{b_j - u}{v - u} f(v), & v - b_j \leq \delta, \\ f(b_j) - f(u) + h_j(b_j - u), & v - b_j > \delta. \end{cases} \quad (47)$$

At a selected cap the branch loss is zero, including the boundary $b_j = u$. The last selected level u carries tail cost

$$T(u) = \sum_{j: b_j \geq u} \pi_j \{f(b_j) - f(u) + h_j(b_j - u)\}. \quad (48)$$

No original-cap anchoring is imposed on catalog entries.

Theorem 12.1 (Global catalog frontier with level-dependent charges). *For strictly increasing concave f and convex nondecreasing h_j , the following dynamic program globally minimizes service loss plus selected-level*

charges under the bounded-overflow institution:

$$\begin{aligned}
 H_1(v) &= \begin{cases} \rho(v), & v \leq b_1, \\ +\infty, & v > b_1, \end{cases} \\
 H_s(v) &= \rho(v) + \min_{u \in \mathcal{A}: u < v} \{H_{s-1}(u) + K_\delta(u, v)\}, \quad s \geq 2, \\
 W_m^\delta &= \min_{1 \leq s \leq \min(m, N), v \in \mathcal{A}} \{H_s(v) + T(v)\}.
 \end{aligned} \tag{49}$$

Backtracking recovers the book, intermediate tiers, and lottery probabilities. All budgets through m require $O(kN^2 + mN^2)$ arithmetic/evaluation work with a direct branch-sum precomputation, and $O(N^2 + mN)$ storage. For rational quadratic data and rational catalog charges the calculation is exact in rational arithmetic. An arbitrary additional charge depending only on s can be added at the final minimization, but level charges must enter the recurrence itself.

Proof. For a fixed book, Proposition EC.5.1 gives the optimal branch rule. On $u \leq b_j < v$, if $v - b_j \leq \delta$ use the adjacent unbiased lottery of mean b_j and set $y_j = 0$; otherwise use u deterministically and set $y_j = b_j - u$. Branches above the last level also use that last level deterministically. Consequently the book's exact loss is the sum of its adjacent-edge costs and its terminal tail. Every branch is counted exactly once; cap endpoints contribute zero regardless of the interval convention.

Increasing paths through the catalog enumerate every possible sorted book. A path can begin exactly at levels no larger than b_1 , which is necessary and sufficient for feasibility. Charging ρ at each visited vertex adds the book's actual implementation cost. Induction on path length gives (49); appending the tail and minimizing over feasible lengths proves both inequalities needed for global optimality. If a selected level is unused, deleting it weakly lowers its nonnegative charge and does not worsen service. Thus the at-most-budget optimization also covers redundant symbols. Store predecessor indices and replay the fixed-book rule to reconstruct a controller.

There are $O(N^2)$ edges, each computable by k branch evaluations, and m layers of $O(N^2)$ comparisons/additions. Storing edges and backpointers gives the stated bounds. In the quadratic rational model every interpolation, difference, and cost is rational. There is no continuous optimizer or rounding tolerance in the recurrence. At $\delta = 0$ every strictly bracketing upper level is excluded, so the branch rule is the deterministic one; at $\delta \geq 1$ every possible upper level is admissible. The endpoint claims follow after optimization over the same catalog. $\square$

This theorem solves the saturated catalog specialization of the unified design problem for arbitrary k, N, m, δ and nonuniform ρ . An exact catalog optimum is not itself an optimum over all real codewords; Theorem 11.1 supplies the missing quantitative comparison. This recurrence does not assume that risk-limited costs are Monge. The continuous uncharged expected frontier retains its separate $O(mk)$ guarantee; the certified-catalog theorem uses a general ordered-path algorithm to accommodate actual level charges and risk eligibility. Entropy coding, shared-table compression depending on pairs of levels, repeated-customer state, and arbitrary codebook-dependent costs are not silently reduced to additive ρ .

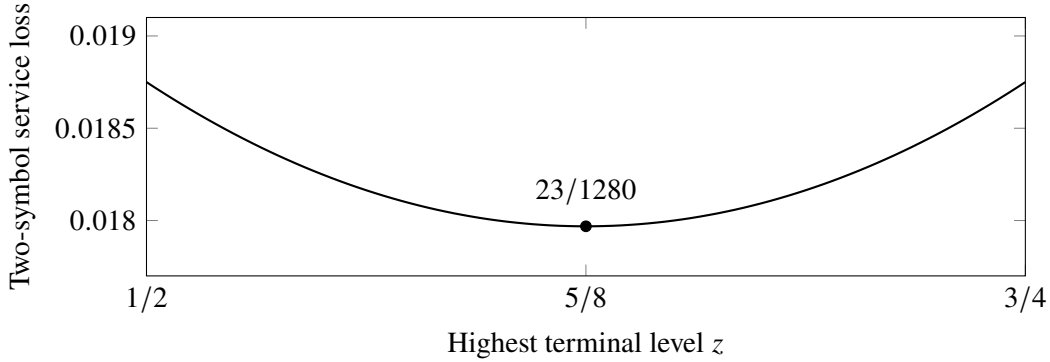


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is $1/4$. The displayed function is $z^2/20 - z/16 + 3/80$ on $[1/2, 3/4]$; its interior minimum is $z = 5/8$, not a branch cap.

Why a loss frontier followed by a resource surcharge is insufficient. Use the off-cap model of Proposition 6.3 and catalog $\{1/4, 1/2, 5/8, 3/4\}$ with two symbols and $\delta = 1$. With zero charges, $(1/4, 5/8)$ attains loss $23/1280$. Give only level $5/8$ charge $1/640$. That book now costs $25/1280$, whereas each cap-only pair $(1/4, 1/2)$ and $(1/4, 3/4)$ costs $24/1280$. The optimum changes its selected levels, not merely its capacity. The exact catalog recurrence finds the change; retaining the uncharged loss-minimizing book and adding its storage bill does not.

Risk exposure along the continuous example. The original two-symbol continuous example is retained as an exact illustration beyond a fixed catalog. Its globally optimal top level and loss are

$$z_\delta = 1/2 + \min(\delta, 1/8), \quad D_2^\delta = z_\delta^2/20 - z_\delta/16 + 3/80. \quad (50)$$

On the middle branch the high-realization probability is

$$p_{\text{high}}(\delta) = \frac{1/4}{z_\delta - 1/4}, \quad \text{maximum overrun} = z_\delta - 1/2.$$

At zero tolerance this branch uses its cap deterministically; at tolerance $1/8$, the high probability is $2/3$ and the possible overrun is $1/8$. Thus probability, mean obligation, and maximum overrun are separate reported quantities. Figure 1 shows the continuous loss profile; the complete global-example proof is in the electronic companion.

13 Pricing Writable Capacity and Other Implementation Resources

Writable state, read-only constants, evaluation work, and compilation are different resources. A finite menu of implementations t for institution I , with charges depending only on capacity m , can be compared by

$$\min_{I,m,t} \{D_m^I + \lambda_w \lceil \log_2 m \rceil + \lambda_R R_{I,t}(m) + \lambda_T Q_{I,t}(m) + C_{I,t}(m)/H_0 + \kappa_I\}. \quad (51)$$

Here H_0 is the number of repeated deployments over which compilation is amortized, and κ_I is an exogenous protocol charge. This is accounting followed by finite comparison, not an additional structural optimization theorem. Its validity requires the stipulated menu charges to be independent of the particular book at a given capacity. Theorem 12.1, in contrast, puts actual additive level charges inside book selection.

With only a writable-bit price, capacity 2^w uses the frontier at $\min(2^w, k)$. For the equal-probability three-cap example, the zero-bit loss is $7/16$, the one-bit losses are $1/96$ under expected and $1/8$ under pathwise participation, and two bits attain full information. Thus a bit price strictly between $1/96$ and $1/8$ selects one bit under expected participation and two under pathwise participation when all other protocol charges agree. A different exogenous κ_I can change that comparison. This example interprets the proved frontiers; it does not estimate the monetary value of certification or customers' tolerance for risk.

14 Inherited Verification and Computational Evidence

The computational package separates exact finite checks, practical measurements, and a standard-quantizer comparison. The theorems establish the all-size and continuous claims; finite checks are regression evidence. Prices and branch populations are synthetic inputs, not empirical estimates.

Preceding promise and catalog checks. Independent enumeration of all catalog subsets and all one/two-point branch supports agrees with every reported finite-catalog frontier, without assuming adjacent support in the comparison oracle. Tests include repeated caps, noncap catalog entries, zero intermediate curvature, nonuniform charges, and zero, intermediate, and unrestricted overrun tolerances. Clipped allocations satisfy exact weighted promises and allocation optimality conditions; direct scaling and saturation lifts verify the value inequalities on heterogeneous policies. Fixed-book responses are checked for concavity across their kinks. Invalid floating-point and malformed inputs are rejected.

The executed suite records 768 catalog frontier equalities, 26,604 direct-support book checks, 600 scaling/lift checks, 660 clipped-allocation checks, 2,460 response-concavity checks, and 128 repeated-cap equalities. The charge-switch example and the interval in (24) are checked in exact fractions.

Comparison with an ordinary scalar quantizer. The baseline recursion over ordered cells is the same recursion used for scalar mean-square quantization; the cell oracle is not. For the three equal-probability caps, both optimal two-cell mean-square partitions have loss $1/96$. Their centroids are $(1/4, 5/8)$ or $(3/8, 3/4)$, respectively. Each merged cell includes a branch whose cap is below its centroid, so that centroid reconstruction is infeasible under the pathwise contract, even with nonnegative intermediate service. This exact comparison identifies the changed objective and constraint rather than benchmarking two algorithms that solve different problems. The expected contract instead uses the classical unbiased encoder with a contractual tail.

Inherited continuous-frontier evidence. The complete preceding exact suite, including completion ties, selected all-padded rows, controller replay, independent PADS substitution, and earlier regression suites, is

rerun in isolated directories. Historical source and result files are not overwritten. The preceding process-isolated continuous-frontier timings through 16,384 branches and reduced-network HiGHS certificates are retained with their original provenance in the companion. The HiGHS comparison validates the finite reduction, not the cap-anchoring theorem or performance at the larger scale. PADS substitutes the search layer only. Neither comparison is presented as a new experiment on operational data.

Additional catalog landscapes. The executed catalog study uses uniform caps, two separated cap clusters, and a population with concentrated branch weights. Each family is evaluated at $(k, N) = (32, 16)$ and $(128, 32)$, eight-symbol capacity, and tolerances 0, $1/8$, 1, with heterogeneous curvatures and level prices. All 18 cases reconstruct and replay their selected books exactly. Times include catalog construction inside the solver, the dynamic program, reconstruction, and direct replay in one process; they are not directly compared with the predecessor’s process-isolated timings. Full numerical records and the timing definition accompany the code.

The nonsaturated extension additionally passes 1715 exact primal/dual-event comparisons, 6955 supporting branch maxima, 72 saturated catalog/DP equalities, 48 interior-promise catalog comparisons, and 480 arbitrary-promise transport checks. The supporting-price equality certifies each fixed-book solution; finite regression is not used as proof of the general theorem.

15 Decomposition, Global Bounds, and Independent Computation

The experiments separate exact mathematical bounds from numerical corroboration and source provenance. Every input is synthetic and every reported run, including a stopped run, is retained. The preceding continuous promise–risk–price study, its exact optima, and its tables are reproduced in the electronic companion, Section EC.11; none of that evidence is deleted or relabeled as a new run.

Exact independent comparisons. The seeded test suite uses 72 models with heterogeneous costs, repeated caps, zero and saturated promises, and both monotone and nonmonotone charges. An independent explicit-lottery oracle checks 432 priced root problems and 6,180 prefix-completion problems against exhaustive catalog search. All 72 completed prefix searches equal the independent catalog enumeration. A further 72 deliberately interrupted runs enclose the exhaustive optimum, and 72 altered upper-bound records are rejected. Five exact uniform-grid comparisons reproduce Proposition 11.3. These checks test implementations and witnesses; the proofs establish the claims for all inputs.

Variable branch count and alphabet budget. Table 2 reports the new algorithm, not the saturated Monge implementation. Its largest branch count is 256; budgets reach 8 and catalogs reach 65 levels. Caps are distinct and costs are heterogeneous, so repeated-cap aggregation is not the explanation for scaling. The declared limit is 40 expanded prefixes per instance, and all bounds are independently checked. All 12 of the 12 reported catalog searches close their exact optimization gap within that limit. This is an observation about the stated family, not a worst-case efficiency theorem. The residual continuous gap includes the full mesh

term even when catalog optimization is exact. Detailed rows in the companion vary the intermediate-cost scale and the charge slope, and record evaluated books, traced memory, and rational bit lengths. The charge is a fixed fee plus a nondecreasing surcharge, not the specially shaped charge of Proposition 9.4.

Enumeration frontier and a different global formulation. The instrumented factorial study completes 19 of 25 continuous face instances and explicitly records the remaining 6 at the 5,000-system budget. It includes four-branch, two-symbol runs and a five-branch case; incomplete face search contributes no purported global upper bound. On 6 separately modeled one- and two-branch instances, SCIP’s original-policy nonlinear formulation produces numerical intervals consistent with every completed rational optimum (6 comparisons). Both book sizes terminate with solver status optimal on 6 of these 6 instances; all statuses and residual intervals are reported, including any time limits. Section EC.10 gives the original probability-and-service formulation, which shares no cell-generator or price-DP code. Its floating-point bounds are corroboration within stated tolerances, not rational certificates. The prefix checker supplies the latter for the catalog problem.

16 Conclusion

Limited execution memory creates a joint allocation and alphabet-design problem: participation and re-alization limits determine which terminal lotteries implement an accepted obligation. At saturation, the contractual tail selects an off-cap highest level, and its crossing-difference identity enables classical linear-time matrix search after continuous minimization. Away from saturation, the eligibility-prefix reduction preserves endogenous branch promises, heterogeneous intermediate costs, and actual selected-level charges in one formulation.

The price decomposition turns that geometry into an algorithmic advantage. Its ordered path recurrence solves each relaxed shared-book problem in polynomial work with variable branch count and alphabet budget, and gives exact completion bounds for prefix search. The resulting global interval is backed by a feasible lottery and a checkable exhaustive partition, not merely by source provenance or an inner multiplier. A uniform rounding theorem transfers even an interrupted catalog search to continuous bounds, and a fixed-input construction proves its mesh rate sharp.

The general concave model has exact fixed-book structure, a scalar-oracle price decomposition, and feasible discretization bounds; exact unrestricted continuous optimization is established for the rational quadratic subclass. The continuous face algorithm remains an exponential reference, and global prefix search also has exponential worst-case size. The new polynomial subproblem and verified completion bounds are not claims of polynomial global optimization or an FPTAS. These distinctions identify the contribution’s scope without replacing the original continuous-design result.

The retained phase diagram, institutional-gap bounds, repeated-cap aggregation, and robustness of off-cap design describe concrete decisions within the stipulated model. The additional monotone-charge theorem shows why a one-branch pathwise-randomization separation needs the deliberately nonmonotone charge in the retained example. The abstract ordered-interface theorem makes the method reusable independently of

the operational narrative; the synthetic experiments test the mathematics and algorithms rather than claiming a calibrated deployment. Post-draw exit, hidden reports, coupled opening charges, and additional history channels remain different model specifications, not tacit consequences of this one.

Data and Code Availability

The accompanying repository provides the rational price decomposition, prefix search, independent global-interval checker, original-policy SCIP formulation, instrumented continuous reference solver, seeded synthetic inputs, raw results, and complete certificate ledgers under `revisions/or-r43-prefix-decomposition-20260924`. Its README gives executable reproduction commands and identifies the immutable review baseline. Prior revisions, derivations, proofs, reports, and reader snapshots are preserved. No proprietary observations or customer data are used; the institutional model and its parameters are stated assumptions, not an empirical calibration. The build-validation record reports the committed source identity separately from mathematical and computational verification.

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Table 1: Theorem-level comparison with established principles.

Result here	Predecessor principle	Contract-specific statement proved here
Deterministic cells	Ordered quantizer DP; Wu	Participation fixes lower-cap reconstruction with heterogeneous intermediate costs.
Adjacent lottery	Stochastic rounding; dual quantization	Expected participation selects that primitive jointly with shortfall service; pathwise participation changes feasibility.
Continuous top level	Optimized quantizer grids; Jourdain–Pagès	Cap-anchored representation combined with a heterogeneous, one-sided contractual tail; exact off-cap optimum.
Linear frontier	Monge DP and classical matrix search	Tail crossing difference, optimized Monge order, and exact oracle/domain verification make the classical bound applicable.
Promise robustness	Concave allocation and Lipschitz comparison	Same-alphabet transport gives all-promise bounds, exact fixed-book certificates, and a strict nonsaturated advantage.
Charged risk catalog	Ordered shortest paths; fixed-opening costs	Exact risk-eligibility edge costs and actual level charges jointly optimize service and implementation.
Joint continuous design	Classical rational quadratic optimization	Eligibility-prefix and closed-cell elimination solve endogenous promises, risk and level charges together.
Certified mesh design	Quantizer approximation and rounding	Downward rounding plus probability repair preserves every target and ceiling, with a uniform additive certificate.

Table 2: New prefix search and continuous enclosures. Each row has a separately checked global catalog certificate.

Case	k	N	m	L	U	$K_m h$	Seconds
0	8	17	2	0.620976	0.620977	0.205625	0.41
1	8	17	4	0.625546	0.625547	0.208125	0.32
2	8	33	4	0.628203	0.628204	0.104063	3.71
3	32	17	2	0.639086	0.639087	0.205625	1.56
4	32	33	4	0.645389	0.645390	0.104063	4.50
5	128	33	4	0.651343	0.651344	0.104063	28.65
6	128	33	8	0.651343	0.651344	0.106563	28.67
7	256	33	8	0.654397	0.654398	0.106563	211.40
8	32	33	4	0.658514	0.658515	0.062500	3.47
9	32	33	4	0.645389	0.645390	0.221250	18.48
10	32	33	4	0.529522	0.529523	0.126563	12.59
11	32	65	4	0.645414	0.645415	0.052032	25.08

L is rounded downward; U and $K_m h$ are rounded upward. The exact rational catalog gap is zero in each displayed row, though outward rounding can separate the printed endpoints. The continuous interval is $[L, U + K_m h]$. Detailed parameters and separate checker timings appear in Table EC.11. Seconds are one instrumented run with allocation tracing, not averages or uninstrumented speed comparisons.